A numerically stable edge sum for the Fraunhofer diffraction factor of a polygonal aperture

GOAD theory notes

1 Motivation

GOAD maps each near-field beam to the far field by evaluating a Fraunhofer diffraction integral over the beam's polygonal aperture. The implementation in `diff2.rs` reduces this integral to a sum over the polygon edges. Each edge term in the current form contains the factor $1/(pq)$, where p and q are the transverse components of the scattering wavevector mismatch. The factors are singular whenever the observation direction approaches the beam propagation direction in either transverse coordinate. The code guards these singularities by clamping $|p|$ and $|q|$ to a floor value (`KXY_EPSILON` $= 10^{-3}$).

The clamp avoids division by zero but introduces two artefacts. First, the result is frozen to a constant wrong value inside the clamp band, since the formula no longer responds to the true p or q. Second, the result jumps discontinuously when p or q crosses the clamp boundary. Backscattering is particularly sensitive: the retro-reflected beams that dominate the $\theta = 180°$ bin all have small p and q, and the backscatter Mueller element S_{11} is a strongly cancelling coherent sum, which amplifies per-beam errors of order 0.1% into errors of several percent. This is the numerical reason behind the `ZONE_THETA_OFFSET` guard, which places the forward and backward zone bins at $0.01°$ away from the exact poles.

This note derives an equivalent edge sum in which the singularity never appears. The result is exact at $p = q = 0$, smooth through the poles, and removes the need for `KXY_EPSILON`, the slope guards `m_adj` and `n_adj`, and the `ZONE_THETA_OFFSET` workaround.

2 Setting and notation

Work in the aperture frame, where the beam cross section is a planar polygon in the xy plane with vertices $v_j = (x_j, y_j)$ for $j = 0, \ldots, N-1$, taken cyclically so that $v_N = v_0$. Write the edge vectors as

$$\Delta x_j = x_{j+1} - x_j, \quad \Delta y_j = y_{j+1} - y_j. \tag{1}$$

The far-field contribution of the beam at observation direction $\hat{k}$ factorises into an amplitude part and the scalar Fraunhofer factor. The Fraunhofer factor is proportional to the two-dimensional Fourier transform of the aperture indicator function,

$$A(p, q) = \iint_A e^{i(px+qy)} \, \mathrm{d}x \, \mathrm{d}y, \tag{2}$$

evaluated at the wavevector mismatch

$$p = k\big(\hat{k}_{\mathrm{inc}}, x - \hat{k}_x\big), \quad q = k\big(\hat{k}_{\mathrm{inc}}, y - \hat{k}_y\big), \tag{3}$$

where k is the wavenumber and $\hat{k}_{\mathrm{inc}}$ is the beam propagation direction in the aperture frame. In the code, p and q are `kxx` and `kyy`. The full far-field contribution is

$$E\big(\hat{k}\big) \propto -i \frac{k}{2\pi} e^{ik\hat{k}\cdot r_0} A(p, q), \tag{4}$$

where r_0 is the aperture centre of mass in the lab frame. The prefactor and phase correspond to `inv_denom` and `bvsk` in the code and are unchanged by everything below. The task is to evaluate $A(p,q)$ stably for all p, q, including the limit $p, q \to 0$ where $A \to$ Area.

3 The current edge sum and its singular structure

Let $m_j = \Delta y_j / \Delta x_j$ be the edge slope and define

$$\delta_j = p x_j + q y_j, \quad \delta_{1,j} = p + m_j q, \quad \omega_j = \Delta x_j \delta_{1,j} = p \Delta x_j + q \Delta y_j. \tag{5}$$

The implementation accumulates, per edge, a term algebraically equal to

$$F_j = -i e^{i\delta_j} (e^{i\omega_j} - 1) \frac{p - m_j q}{2pq\delta_{1,j}}, \tag{6}$$

and the total is $\sum_j F_j = -iA(p,q)$. The factor $1/(pq)$ in every term is the problem: each term diverges as $p \to 0$ or $q \to 0$, even though the sum remains finite because the divergences cancel between edges. The cancellation is exact in real arithmetic but is destroyed by the clamps, and is badly conditioned in `f32` even without them.

The origin of the double singularity is that Equation 6 is the average of two distinct exact representations of A, as shown next. Averaging puts both $1/p$ and $1/q$ into every term. Using either representation alone, chosen appropriately, removes the smaller denominator entirely.

4 Two exact single-coordinate forms

Green's theorem in the plane states, for differentiable L and M,

$$\iint_A \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} \right) \mathrm{d}x \, \mathrm{d}y = \oint (L \, \mathrm{d}x + M \, \mathrm{d}y), \tag{7}$$

with the contour traversed in the positive (counterclockwise) sense. Two choices of potential reproduce the integrand $e^{i(px+qy)}$:

1. $L = 0$, $M = e^{i(px+qy)}/(ip)$, giving

$$A(p,q) = \frac{1}{ip} \oint e^{i(px+qy)} \, \mathrm{d}y. \tag{8}$$

2. $M = 0$, $L = -e^{i(px+qy)}/(iq)$, giving

$$A(p,q) = -\frac{1}{iq} \oint e^{i(px+qy)} \, \mathrm{d}x. \tag{9}$$

Note that Equation 8 contains no $1/q$ and Equation 9 contains no $1/p$.

On the straight edge from v_j to v_{j+1}, parameterise $r(t) = v_j + t\big(\Delta x_j, \Delta y_j\big)$ with $t \in [0,1]$. Then $px + qy = \delta_j + t\omega_j$ and $\mathrm{d}y = \Delta y_j \, \mathrm{d}t$, so

$$\int_{\text{edge } j} e^{i(px+qy)} \, \mathrm{d}y = \Delta y_j e^{i\delta_j} \int_0^1 e^{it\omega_j} \, \mathrm{d}t = \Delta y_j e^{i\delta_j} \varphi\big(\omega_j\big), \tag{10}$$

with the entire function

$$\varphi(\omega) = \frac{e^{i\omega} - 1}{i\omega}, \quad \varphi(0) = 1. \tag{11}$$

The two stable edge sums are therefore

$$A(p,q) = \frac{1}{ip}\sum_{j=0}^{N-1} \Delta y_j e^{i\delta_j}\varphi(\omega_j), \tag{12}$$

$$A(p,q) = -\frac{1}{iq}\sum_{j=0}^{N-1} \Delta x_j e^{i\delta_j}\varphi(\omega_j). \tag{13}$$

Both are exact for all p, q where they are defined. Only δ_j and ω_j are needed per edge. No slopes appear, so the guards on near-vertical and near-horizontal edges (`m_adj`, `n_adj`, `DIFF_DMIN`) are unnecessary in this formulation.

Averaging Equation 12 and Equation 13 and substituting $\Delta y_j = m_j \Delta x_j$ recovers Equation 6 after a few lines of algebra. This confirms the equivalence with the current implementation and shows that the $1/(pq)$ structure is an artefact of the symmetrisation, not intrinsic to the problem.

5 Evaluation of φ for small argument

φ is entire, with Taylor series

$$\varphi(\omega) = 1 + \frac{i\omega}{2} - \frac{\omega^2}{6} - \frac{i\omega^3}{24} + \mathcal{O}(\omega^4). \tag{14}$$

For $|\omega| < \omega_0 = 10^{-2}$ the truncation error of Equation 14 is below $\omega_0^4/120 \approx 8 \times 10^{-11}$, far below `f32` resolution, so the series can be used there and the closed form $\varphi(\omega) = \sin(\omega)/\omega + i(1 - \cos(\omega))/\omega$ elsewhere. This removes the remaining removable singularity at $\omega_j = 0$.

6 The limit of small p and q

When both $|p|$ and $|q|$ are small the prefactor $1/(ip)$ in Equation 12 still amplifies rounding error, because the leading terms of the sum cancel. The limit can instead be taken analytically. Expand $e^{i\delta_j}\varphi(\omega_j)$ to first order in the small quantities δ_j and ω_j and insert into Equation 12:

$$A = \frac{1}{ip}\sum_j \Delta y_j \left(1 + i\delta_j + \frac{i\omega_j}{2} + \mathcal{O}(2)\right). \tag{15}$$

Three elementary identities for a closed polygon evaluate the sums:

1. $\sum_j \Delta y_j = 0$, by telescoping.
2. $\sum_j \Delta y_j \left(y_j + \Delta\frac{y_j}{2}\right) = \frac{1}{2}\sum_j \left(y_{j+1}^2 - y_j^2\right) = 0$, by telescoping.
3. $\sum_j \Delta y_j \left(x_j + \Delta\frac{x_j}{2}\right) = \oint x\,\mathrm{d}y = S$, the signed area, exact for straight edges.

Collecting terms, with $\delta_j + \omega_j/2 = p\left(x_j + \Delta x_j/2\right) + q\left(y_j + \Delta y_j/2\right)$:

$$A(p,q) = S + i\left(pM_x + qM_y\right) + \mathcal{O}\left((\kappa R)^2\right)S, \tag{16}$$

where $\kappa = \max(|p|, |q|)$, R is the maximum vertex distance from the origin, and the first moments are

$$M_x = \iint_A x\,\mathrm{d}A, \quad M_y = \iint_A y\,\mathrm{d}A, \tag{17}$$

given for a polygon by the standard shoelace-type formulas

$$S = \frac{1}{2}\sum_j \left(x_j y_{j+1} - x_{j+1} y_j\right), \tag{18}$$

$$M_x = \frac{1}{6}\sum_j (x_j + x_{j+1})(x_j y_{j+1} - x_{j+1} y_j), \quad M_y = \frac{1}{6}\sum_j (y_j + y_{j+1})(x_j y_{j+1} - x_{j+1} y_j). \tag{19}$$

In GOAD the aperture is translated to its centre of mass before diffraction, so $M_x \approx M_y \approx 0$ and Equation 16 reduces to $A \approx S$. The moment terms are retained anyway since they are cheap and make Equation 16 correct to second order for any polygon origin. Note that S is the signed area. The sign encodes the winding direction of the aperture polygon and must not be replaced by the absolute area, since Equation 12 and Equation 13 carry the same sign convention.

7 Branch selection and error budget

The evaluation strategy is:

1. If $\kappa R < \tau$: use the limit Equation 16. The neglected second-order term gives relative error $\mathcal{O}(\tau^2)$.
2. Otherwise, if $|p| \geq |q|$: use Equation 12, dividing only by the larger component. Rounding in the cancelling sum gives relative error of order $\varepsilon_{\text{f32}} \cdot \mathcal{P}/(|p|\ |S|)$ with $\mathcal{P}$ the perimeter scale, which is largest at the branch boundary $|p| = \tau/R$.
3. Otherwise: use Equation 13.

With $\tau = 3 \times 10^{-3}$ both error sources stay near or below 10^{-4} relative, compared with 1.5×10^{-3} and discontinuous for the clamped form in the small-(p, q) regime. Any τ in $[10^{-3}, 10^{-2}]$ is acceptable; the total error is insensitive to the exact choice.

8 Numerical validation

The formulas were validated against dense `float64` quadrature of Equation 2 on a centred hexagonal aperture of circumradius 3, with the edge sums evaluated in `f32` arithmetic to mimic the Rust implementation. The script is `fraunhofer_limit_check.py` in this directory.

case	clamped form error	stable form error
generic p, q	0.001%	0.0005%
q inside clamp band	0.024%	0.0003%
$q = 0$ exactly	0.036%	0.0003%
both inside clamp band	0.152%	0.00001%
backscatter pole ($p = 2 \times 10^{-5}$, $q = -8 \times 10^{-8}$)	0.152%	$\approx 0\%$
$p = q = 0$ exactly	0.152%	$\approx 0\%$

The clamped form additionally returns the identical value for every (p, q) inside the clamp band and jumps at its boundary. In a full GOAD run on a hexagonal column (`config/local.toml`, seed 6), this produced erratic jumps of -1.8% and $+3.0\%$ in the backscatter S_{11} within $0.0001°$ of $\theta = 180°$, against a smooth extrapolated limit. The stable form removes the discontinuity, which makes evaluation at exactly $\theta = 0°$ and $\theta = 180°$ legitimate and allows `ZONE_THETA_OFFSET` to be set to zero.

9 Implementation notes

- p, q are the raw `kxx`, `kyy` from `calculate_kxx_kyy` with the clamps removed.
- The per-edge inputs are only δ_j and ω_j; `EdgeData` needs the raw Δx_j, Δy_j (without the `DIFF_DMIN` floor) and no longer needs `m_adj`, `n_adj`.
- The signed area and first moments can be precomputed once per beam in `EdgeData::from_vertices`.

- The final Fraunhofer factor keeps its existing prefactors: multiply $A(p,q)$ by $-i$, by `inv_denom` $= k/(2\pi)$, and by the phase $e^{i\cdot\,\mathrm{bvsk}}$.
- The legacy geometric-optics path in `diff.rs` has its own copy of the clamped formulas and is not affected.

10 Outlook: generalised incident fields

The formalism so far assumes the incident field is a uniform plane wave across the aperture, so that the integrand of Equation 2 carries no profile factor. This section records which deviations from a plane wave preserve the contour reduction. The generalised factor is

$$A_U(p,q) = \iint_A U(x,y) e^{i(px+qy)}\,\mathrm{d}x\,\mathrm{d}y, \tag{20}$$

where U is the complex field profile on the aperture.

10.1 The structural criterion

The edge-sum derivation uses exactly two operations to produce closed-form terms: an antiderivative of the integrand in one Cartesian variable, to build the Green's theorem potential, and integration of that potential along straight edges. Both stay elementary whenever the integrand of Equation 20 belongs to the class of exponential polynomials, meaning finite sums of

$$(\text{polynomial in } x,y) \times e^{\text{linear form in } x,y} \tag{21}$$

with complex coefficients. This class is closed under both operations: antiderivatives of $x^n e^{\alpha x}$ remain in the class, and restriction to a straight edge gives a polynomial times an exponential in the edge parameter t. The question of whether the contour integral survives therefore reduces to whether U can be written, or well approximated, in this class.

10.2 Cases

Polynomial amplitude profile. Exact. For a linear profile $U = a + bx + cy$,

$$A_U = aA_0 - ib\frac{\partial A_0}{\partial p} - ic\frac{\partial A_0}{\partial q}, \tag{22}$$

where A_0 is the plane-wave edge sum Equation 12. The edge terms are closed-form in p and q, so their derivatives are too; the result is a second edge sum with modified per-edge weights. Higher polynomial orders follow from higher derivatives. In the small-argument limit the area is replaced by the correspondingly weighted moments.

Linear phase error, tilted wavefront, or linear absorption gradient. Exact by argument shift. For $U = e^{\mu x + \nu y}$ with complex μ, ν,

$$A_U(p,q) = A_0(p - i\mu, q - i\nu). \tag{23}$$

The stable form continues analytically since φ is entire. A pure linear phase ramp is a redefinition of the incidence direction, as expected.

Sinusoidal amplitude modulation. Exact with two evaluations: $\cos(g \cdot r)$ splits into two complex exponentials, so A_U is the average of A_0 at $\big(p \pm g_x, q \pm g_y\big)$.

Sinusoidal phase modulation. This is a thin phase grating, handled by the Jacobi-Anger expansion:

$$e^{im\sin(g\cdot r)} = \sum_{n=-\infty}^{\infty} J_n(m) e^{ing\cdot r}, \tag{24}$$

so A_U is a sum of shifted edge-sum evaluations weighted by Bessel functions. The series truncates fast because $J_n(m)$ collapses super-exponentially once $n > m$; for modulation depth $m \sim 1$ about five terms suffice. The terms are the grating orders, which provides a free validation target.

Phase noise. Smooth correlated noise written as a few Fourier modes gives a product of Jacobi-Anger expansions, one per mode. This is workable for a handful of modes and combinatorial beyond that. The general-purpose fallback is subdivision: triangulate the aperture, linearise the phase over each element, and apply the argument-shift case per triangle exactly. The error is controlled by the quadratic remainder per element, so the element count scales with the total phase excursion. Noise that is rough at the wavelength scale is a different regime; there the coherent edge sum is the wrong tool and a statistical treatment (Strehl-type attenuation of the coherent term plus a diffuse pedestal) is more honest.

10.3 Where the reduction breaks

Quadratic phase, meaning wavefront curvature or defocus, is the boundary of the method. The antiderivative of a Gaussian along an edge brings in erf of complex argument, and the subsequent edge integral of erf $\times$ Gaussian is not elementary. Curvature therefore requires either Fresnel-type special functions per edge or subdivision until the phase is locally linear. Everything below quadratic is exact.

10.4 Approach to implementation

A practical route, in order:

1. Extend `polygon_ft` to accept complex (p, q) and add the $\partial/\partial p$, $\partial/\partial q$ variants. Together these cover polynomial amplitude, linear phase, and sinusoidal modulation exactly, with the stability machinery of this note carrying over unchanged.
2. Validate against the same brute-force quadrature harness as `fraunhofer_limit_check.py` with a nonuniform U.
3. Decide how far up the hierarchy is needed. Linear amplitude plus linear phase per beam likely covers most of what beam tracing produces, since beam subdivision already plays the role of piecewise linearisation.

One caveat: a plane wave with a linear amplitude profile is not an exact solution of the Helmholtz equation, so U should be understood as a Kirchhoff-style boundary field on the aperture rather than a global incident field. The near-to-far mapping requires nothing more.